UDP TIME SERVER PROGRAM

SERVER

```
import java.net.*;

public class UDPTimeServer {
   public static void main(String[] args) {
      try {
         // Step 1: Create DatagramSocket on port 7000
         DatagramSocket serverSocket = new DatagramSocket(7000);
         System.out.println("UDP Time Server started...");

         // Step 2: Buffer to receive data
         byte[] receiveBuffer = new byte[1024];

         // Step 3: DatagramPacket to receive data
         DatagramPacket receivePacket =
               new DatagramPacket(receiveBuffer, receiveBuffer.length);

         // Step 4: Receive request from client
         serverSocket.receive(receivePacket);

         String request = new String(
               receivePacket.getData(), 0, receivePacket.getLength());

         System.out.println("Client Request: " + request);

         // Step 5: Get current system time (NO NEW IMPORT)
         String currentTime =
```

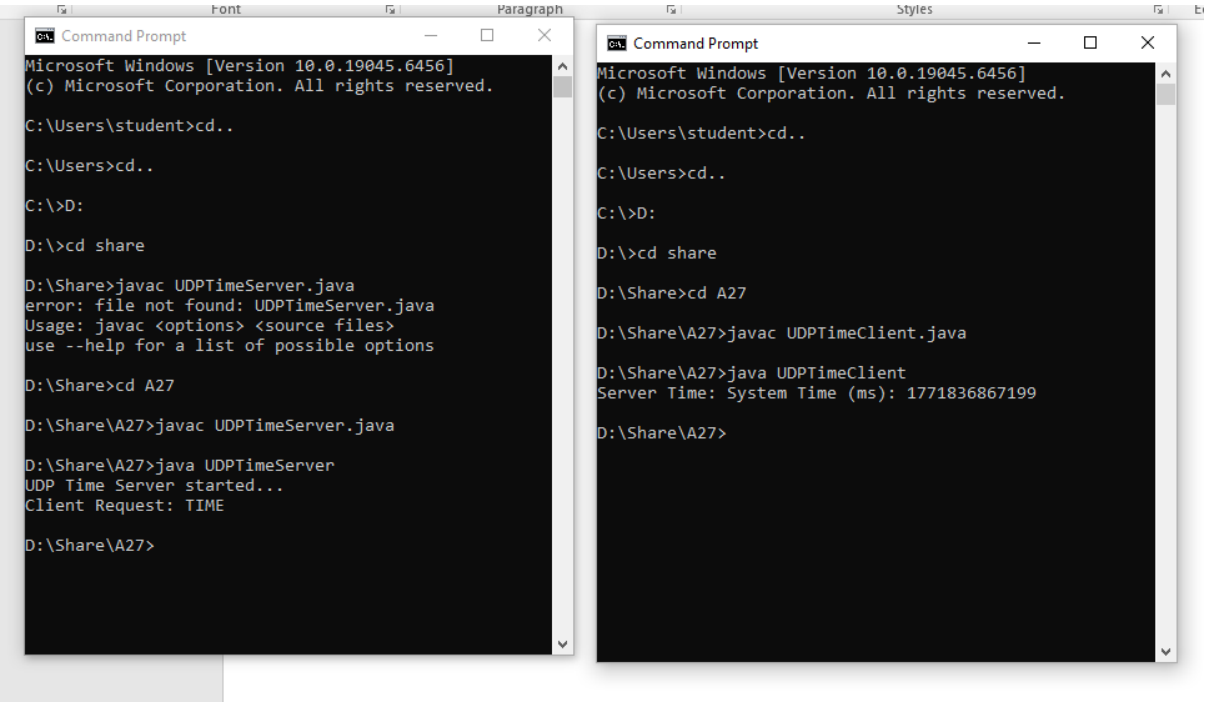
```
            "System Time (ms): " + System.currentTimeMillis();

        byte[] sendBuffer = currentTime.getBytes();

        InetAddress clientAddress = receivePacket.getAddress();
        int clientPort = receivePacket.getPort();

        // Step 6: Send time to client
        DatagramPacket sendPacket =
            new DatagramPacket(sendBuffer, sendBuffer.length,
                clientAddress, clientPort);

        serverSocket.send(sendPacket);

        serverSocket.close();

    } catch (Exception e) {
        e.printStackTrace();
    }
  }
}
```

CLIENT

```
import java.net.*;

public class UDPTimeClient {
  public static void main(String[] args) {
    try {
        // Step 1: Create DatagramSocket
```

```
    DatagramSocket clientSocket = new DatagramSocket();

    // Step 2: Send time request
    String request = "TIME";
    byte[] sendBuffer = request.getBytes();

    InetAddress serverAddress =
        InetAddress.getByName("localhost");

    DatagramPacket sendPacket =
        new DatagramPacket(sendBuffer, sendBuffer.length,
            serverAddress, 7000);

    clientSocket.send(sendPacket);

    // Step 3: Receive time from server
    byte[] receiveBuffer = new byte[1024];
    DatagramPacket receivePacket =
        new DatagramPacket(receiveBuffer, receiveBuffer.length);

    clientSocket.receive(receivePacket);

    String serverTime = new String(
        receivePacket.getData(), 0, receivePacket.getLength());

    System.out.println("Server Time: " + serverTime);

    clientSocket.close();

} catch (Exception e) {
    e.printStackTrace();
```

```
        }
    }
}
```


Font
Paragraph
Styles
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.
C:\Users\student>cd..
C:\Users>cd..
C:\>D:
D:\>cd share
D:\Share>javac UDPTimeServer.java
error: file not found: UDPTimeServer.java
Usage: javac <options> <source files>
use --help for a list of possible options
D:\Share>cd A27
D:\Share\A27>javac UDPTimeServer.java
D:\Share\A27>java UDPTimeServer
UDP Time Server started...
Client Request: TIME
D:\Share\A27>
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.
C:\Users\student>cd..
C:\Users>cd..
C:\>D:
D:\>cd share
D:\Share>cd A27
D:\Share\A27>javac UDPTimeClient.java
D:\Share\A27>java UDPTimeClient
Server Time: System Time (ms): 1771836867199
D:\Share\A27>